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Java Developer – **Toughest Technical** Interview Questions (4–6 YOE)



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Section 1 – Core Java & OOP

1.
Your application runs perfectly in QA, but in Production the JVM suddenly starts consuming 95% CPU after handling 20,000 concurrent requests. How would you investigate?
2.
Two threads update the same employee salary simultaneously. Sometimes incorrect salary is stored.
How would you fix this?
3.
You inherit legacy code with 5000 lines inside one Java class.
How would you refactor it without breaking production?
4.
A HashMap contains 20 million entries.
Searching has become slow.
What could be wrong?
5.
How would you implement a thread-safe Singleton in Java?
Which implementation is best for production?
6.
Your application throws

OutOfMemoryError: Java Heap Space

How would you debug it?
7.
A developer replaced ArrayList with LinkedList.
Application became slower.
Why?
8.
When would ConcurrentHashMap still fail to produce correct business logic?
9.
How would you implement your own BlockingQueue?
10.
Design an immutable Employee class.
How do you make it truly immutable?

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Section 1 – Core Java & OOP

11.

A Comparator gives inconsistent sorting.

How would you debug?

12.

How would you detect a deadlock in production?

13.

How do weak references help in caching?

14.

A Java Stream API implementation is slower than for-loop.

Why?

15.

Explain a production issue caused by incorrect equals() and hashCode() implementation.

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Section 2 – Spring Boot & Spring Internals

16.
Your Spring Boot application startup time increased from 20 seconds to 4 minutes.
Where do you start debugging?
17.
Application suddenly starts returning HTTP 500 for random APIs.
Logs show no exceptions.
How do you troubleshoot?
18.
A singleton bean stores user-specific information.
What production issue can happen?
19.
How would you implement feature toggles without redeployment?
20.
A Spring Boot application consumes 4GB RAM after deployment.
How do you identify the root cause?
21.
How do you implement graceful shutdown in Spring Boot?
22.
How would you secure internal microservice APIs?
23.
What happens internally when Spring Boot starts?
24.
How would you customize Spring Boot AutoConfiguration?
25.
How would you trace a request across 15 microservices?
26.
How do you detect circular dependency between beans?
27.
One endpoint suddenly becomes slow only in production.
What would you investigate?
28.
How would you implement API versioning?
29.
How would you implement global exception handling?
30.
Spring Context initialization fails randomly.
How do you debug?

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Section 3 – Spring Security

31.
JWT token expires while user is performing payment.
How should the application behave?
32.
How do you invalidate JWT before expiry?
33.
A REST API is vulnerable to unauthorized access.
How do you secure it?
34.
How would you implement Role-Based Access Control?
35.
How do you secure communication between microservices?
36.
OAuth server becomes unavailable.
What happens?
37.
Refresh token gets stolen.
How do you minimize damage?
38.
A user changes password.
How do you invalidate existing sessions?
39.
How do you protect APIs from brute-force attacks?
40.
How would you implement API throttling?

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Section 4 – Microservices

41.
Payment Service becomes unavailable.
Order Service should continue working.
How would you design this?
42.
Inventory update succeeds.
Payment fails.
How do you maintain consistency?
43.
How do you avoid cascading failures?
44.
Design Order, Inventory, Payment, Shipping microservices.
How do they communicate?
45.
How would you migrate Monolith to Microservices?
46.
What happens if one service has a different API version?
47.
How do you implement Saga Pattern?
48.
Compensation transaction also fails.
Now what?
49.
How do you prevent duplicate order creation?
50.
Design a highly available authentication service.
51.
How would you detect distributed deadlocks?
52.
One microservice receives 50k requests/sec.
How do you scale?
53.
How would you perform zero downtime deployment?
54.
How do you monitor microservices?
55.
How do you debug intermittent failures across services?

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Section 5 – Hibernate / JPA

56.

Application suddenly starts executing 300 SQL queries for one API.

Why?

57.

Explain a production issue caused by LazyInitializationException.

58.

How would you optimize N+1 queries?

59.

Bulk update takes 20 minutes.

How do you optimize?

60.

Second-level cache returns stale data.

How do you fix?

61.

Database deadlocks occur every day.

How do you investigate?

62.

How would you implement optimistic locking?

63.

When would pessimistic locking be better?

64.

How do you reduce Hibernate memory consumption?

65.

How do you debug slow SQL generated by Hibernate?

Section 6 – SQL & NoSQL (10 Questions)

66.

A query takes:

- Dev → 40 ms
- Production → 9 sec

How do you investigate?

67.

Database CPU reaches 95%.

What are your first five actions?

68.

Users complain about duplicate records.

How do you prevent duplicates?

69.

When would you prefer MongoDB over PostgreSQL?

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Section 5 – Hibernate / JPA

- 70.
How do you design an audit logging database?
- 71.
How would you shard customer data?
- 72.
How do you archive historical data?
- 73.
How do you optimize joins on huge tables?
- 74.
Design a schema for Employee Management System.
- 75.
How would you implement pagination efficiently?

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Section 7 – Kafka / RabbitMQ

76.

Kafka consumer processes duplicate messages.

How do you fix?

77.

Consumer lag reaches 2 million.

How do you investigate?

78.

Producer sends duplicate events.

How do you prevent duplicate business processing?

79.

RabbitMQ queue grows indefinitely.

Why?

80.

How do you retry failed messages safely?

81.

Dead Letter Queue suddenly contains 100,000 messages.

What do you do?

82.

How do you guarantee message ordering?

83.

One partition becomes hotspot.

How do you rebalance?

84.

How do you replay messages safely?

85.

Design an event-driven notification system.

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Section 8 – Docker / Kubernetes / CI-CD

86.

Docker container restarts every few minutes.

How do you debug?

87.

Kubernetes Pod is Running but application is unreachable.

Why?

88.

How do you perform rolling deployment without downtime?

89.

Deployment fails after Jenkins build succeeds.

How do you troubleshoot?

90.

Application works locally but not inside Docker.

Possible reasons?

91.

How do you reduce Docker image size?

92.

Kubernetes pod keeps getting OOMKilled.

How do you fix?

93.

One pod receives all traffic.

Other pods are idle.

Why?

94.

How do you rollback a failed deployment?

95.

How do you secure secrets inside Kubernetes?

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Section 9 – JUnit & Testing

96.

A JUnit test passes locally but fails in Jenkins.

How do you debug?

97.

How do you unit test a service using Mockito?

98.

How would you test asynchronous methods?

99.

How do you test Kafka consumers without a real Kafka cluster?

100.

A production bug was missed even though test coverage is 95%.

How is that possible?

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